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Inventor(s)	Inose; Koji et al.

Vehicle

Abstract

A vehicle that can secure both stable communication by an ECU including a communication unit that unlocks a handlebar lock module and theft prevention. The vehicle includes a steering shaft, a vehicle body frame including a head pipe that turnably supports the steering shaft and a seat rail that is disposed below a seat on which a rider is seated to fix the seat, and a communication unit for communicating with a smart key (a device external to the vehicle). The communication unit is provided below and on the outer side in a vehicle width direction of the seat rail and in a space covered by and on the inner side of a side cover (vehicle body cover) that protects a vehicle body part.

Inventors:	Inose; Koji (Tokyo, JP), Kitamura; Ryohei (Tokyo, JP)
Applicant:	HONDA MOTOR CO., LTD. (Tokyo, JP)
Family ID:	83455988
Assignee:	HONDA MOTOR CO., LTD. (Tokyo, JP)
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Primary Examiner: Barakat; Mohamed

Attorney, Agent or Firm: Rankin, Hill & Clark LLP

Background/Summary

TECHNICAL FIELD

(1) The present invention relates to a vehicle.

BACKGROUND ART

(2) Conventionally, for example, a technique has been disclosed in which an ECU including a vehicle unlocking unit and a communication unit are integrally provided as a control unit in a vehicle (see, for example, Patent Literature 1).

(3) For example, a technique in which a control unit is provided under a rider's seat is disclosed (see, for example, Patent Literature 2).

CITATION LIST

Patent Literature

(4) Patent Literature 1: WO 2020/202343 A Patent Literature 2: JP 2020-111189 A

SUMMARY OF INVENTION

Technical Problem

(5) A conventional technique of providing in a vehicle an ECU including a vehicle unlocking unit for unlocking a handlebar lock module is disclosed. In providing an ECU including a communication unit for unlocking a handlebar lock module in a vehicle, it is required to secure stable communication by a communication unit and theft prevention.

(6) The present invention has been made in view of the above point. An object of the present invention is to provide a vehicle that secures both stable communication by an ECU including a communication unit that unlocks a handlebar lock module and theft prevention.

Solution to Problem

(7) To achieve the above object, in the present invention, a vehicle including a steering shaft, a vehicle body frame including a head pipe that turnably supports the steering shaft, a main frame that extends from the head pipe to a rear side of the vehicle, and a seat rail that is continuous with the main frame and disposed below a seat on which a rider of the vehicle is seated to fix the seat, and a drive device that generates power for the vehicle, includes a communication unit for communicating with a device external to the vehicle, where the communication unit is provided below and on an outer side in a vehicle width direction of the seat rail and in a space covered by and on an inner side of a vehicle body cover that protects a vehicle body part.

(8) This specification includes all the contents of Japanese Patent Application No. 2021-059695 filed on Mar. 31, 2021.

Advantageous Effects of Invention

(9) According to the present invention, since a communication unit is provided below and on the outer side in the vehicle width direction of the seat rail and in a space covered by and on the inner

side of a vehicle body cover that protects vehicle body parts, metal parts do not hinder communication by the communication unit, and stability of communication can be improved. Moreover, providing the communication unit in the space covered by and in the inner side of the vehicle body cover that protects vehicle body parts makes it difficult to remove the communication unit from the vehicle, which secures theft prevention.

Description

BRIEF DESCRIPTION OF DRAWINGS

- (1) FIG. 1 is a perspective view illustrating a configuration of a vehicle according to a present embodiment.
- (2) FIG. 2 is a block diagram illustrating a configuration of a control device according to the present embodiment.
- (3) FIG. 3 is a plan view of a vehicle body according to the present embodiment.
- (4) FIG. 4 is a perspective view of the vehicle body according to the present embodiment as viewed from a right side.
- (5) FIG. 5 is a cross-sectional view taken along line A-A in FIG. 1.
- (6) FIG. 6 is a cross-sectional view illustrating an exemplary modification of the present invention.
- (7) FIG. 7 is a cross-sectional view illustrating another exemplary modification of the present invention.

DESCRIPTION OF EMBODIMENTS

(8) Embodiments of the present invention will be described below with reference to the drawings. Note that, in the description, directions such as front, rear, left, right, up, and down are respectively the same as the directions with respect to a vehicle body unless otherwise specified. In each of the drawings, a reference sign FR denotes the front side of the vehicle body, a reference sign UP denotes the upper side of the vehicle body, and a reference sign LH denotes the left side of the vehicle body.

EMBODIMENTS

- (9) FIG. 1 is a side view illustrating a saddle-ride vehicle **10** according to an embodiment of the present invention.
- (10) The saddle-ride vehicle **10** is a vehicle including a vehicle body frame **11**, a power unit **12** supported by the vehicle body frame **11**, a front fork **14** that supports a front wheel **13** in a steerable manner, a swing arm **16** that supports a rear wheel **15**, and a seat **17** for a rider.
- (11) The saddle-ride vehicle **10** is a vehicle on which the rider is seated astride the seat **17**. The seat **17** is provided in the upper side of a rear part of the vehicle body frame **11**.
- (12) The vehicle body frame **11** includes a head pipe **18** provided at a front end portion of the vehicle body frame **11**, a front frame **19** located rearward of the head pipe **18**, and a rear frame **20** located rearward of the front frame **19**. A front end portion of the front frame **19** is connected to the head pipe **18**.
- (13) The seat **17** is supported by the rear frame **20**.
- (14) The front fork **14** is supported by the head pipe **18** so as to be steerable to the left and the right. The front wheel **13** is supported by an axle **13a**, which is provided at a lower end portion of the front fork **14**. A steering handlebar **21** to be gripped by the rider is attached to an upper end portion of the front fork **14**.
- (15) The swing arm **16** is supported by a pivot shaft **22** supported by the vehicle body frame **11**. The pivot shaft **22** is a shaft extending horizontally in the vehicle width direction. The pivot shaft **22** is inserted through a front end portion of the swing arm **16**. The swing arm **16** swings up and down about the pivot shaft **22**.
- (16) The rear wheel **15** is supported by an axle **15a**, which is provided at a rear end portion of the

swing arm **16**.

(17) The power unit **12** is disposed between the front wheel **13** and the rear wheel **15**, and is supported by the vehicle body frame **11**.

(18) The power unit **12** is an internal combustion engine. The power unit **12** includes a crankcase **23**, and a cylinder portion **24** that houses a reciprocating piston. The cylinder portion **24** has an exhaust port to which an exhaust device **25** is connected.

(19) Output of the power unit **12** is transmitted to the rear wheel **15** by a drive force transmission member that connects the power unit **12** and the rear wheel **15**.

(20) In addition, the saddle-ride vehicle **10** includes a front fender **26** that covers the front wheel **13** from the upper side, a rear fender **27** that covers the rear wheel **15** from the upper side, a step **28** on which the rider places his or her foot, and a fuel tank **29** for storing fuel to be used by the power unit **12**.

(21) The front fender **26** is attached to the front fork **14**. The rear fender **27** and the step **28** are provided below the seat **17**. The fuel tank **29** is supported by the vehicle body frame **11**.

(22) FIG. **2** is a block diagram illustrating a configuration of a control device.

(23) As illustrated in FIG. **2**, a control device **40** includes an ECU **41** as a control unit. The ECU **41** includes, for example, a processor such as a central processing unit (CPU) and a storage unit such as a random access memory (RAM) and a read only memory (ROM).

(24) The storage unit stores data such as an engine control program. Specifically, a processor of the ECU **41** executes a control program to control components of the vehicle.

(25) The ECU **41** is connected to a handlebar lock module **42**. The handlebar lock module **42** is for locking the turning action of the handlebar during parking.

(26) The control device **40** includes a communication unit **44** that receives a signal from a smart key **43** as a device external to the vehicle and carried by the rider.

(27) The control device **40** includes a determination unit **45** that determines whether a signal from the smart key **43** received by the communication unit **44** is a signal for unlocking the handlebar lock module **42**, and an unlocking unit **46** that unlocks the handlebar lock module **42** when determination is made that the signal from the smart key **43** is a signal for unlocking.

(28) By the rider operating the smart key **43**, a signal is transmitted from the smart key **43** to the ECU **41** via the communication unit **44** and the determination unit **45** of the ECU **41** determines whether the signal from the smart key **43** is a signal for unlocking. When the determination unit **45** determines that the signal is for unlocking, the unlocking unit **46** unlocks the handlebar lock module **42**.

(29) FIG. **3** is a plan view of the vehicle body. FIG. **4** is a perspective view of the vehicle body as viewed from the right side. FIG. **5** is a cross-sectional view taken along line A-A in FIG. **1**.

(30) As illustrated in FIGS. **3** and **4**, the rear frame **20** includes a seat rail **30** that supports the seat **17**. Vehicle body parts such as an air cleaner **31** is housed in a space in the upper side of the rear frame **20** and below the seat rail **30**. A side cover **32** as a vehicle body cover is attached on the outer side of the space demarcated by the seat rail **30** and the rear frame **20**.

(31) As illustrated in FIGS. **1** and **2**, an exhaust device **25** as an exhaust mechanism extending rearward and connected to the exhaust port of the power unit **12** is attached on the right side of the vehicle body frame **11**.

(32) A battery **33** as an electric storage device is installed on a side opposite to the right side on which the exhaust device **25** of the vehicle is attached.

(33) In the present embodiment, the control device **40** is installed on a side where the exhaust device **25** is provided and in a space demarcated by the seat rail **30** and the rear frame **20**. In this state, the control device **40** is disposed further on the outer side than the seat rail **30** in the vehicle width direction and in the inner side of the side cover **32**.

(34) With the side cover **32** positioned not to overlap a leg of the rider riding the vehicle in a side view, communication by the communication unit **44** of the control device **40** is not hindered by the

rider when the control device **40** is disposed in the inner side of the side cover **32**.

(35) When the control device **40** is disposed on the inner side of the seat rail **30**, the seat rail **30** made of metal may cause a communication failure. However, in the present embodiment, since the control device **40** is disposed on the outer side of the seat rail **30** and on the side where the exhaust device **25** is installed, that is, on the side opposite to the side where the battery **33** is installed, effects by the seat rail **30** made of metal causing communication failure and the battery **33** can be suppressed, and stable communication by the communication unit **44** of the control device **40** can be secured.

(36) Note that, only the communication unit **44** may be disposed at the place where the control device **40** is disposed. By disposing the communication unit **44** in such a manner, there will be no chance of the communication unit **44** suffering communication failure caused by a metal part such as the seat rail **30**.

(37) In addition, providing the control device **40** on the side opposite to the side where there are many metal parts and where the battery **33** is provided in an easily detachable manner can improve stability of communication, and improve theft prevention without using additional parts.

(38) The control device **40** is fixed to the vehicle body frame **11**. By fixing the control device **40** to the vehicle body frame **11** having high rigidity, assembling work can be performed stably, and work efficiency can be improved.

(39) Next, an effect of the present embodiment will be described.

(40) In the present embodiment, to unlock the handlebar lock module **42**, the rider operates the smart key **43** to transmit a signal from the smart key **43** to the ECU **41** via the communication unit **44**. The determination unit **45** of the ECU **41** determines whether the signal from the smart key **43** is a signal for unlocking, and when the determination unit **45** determines that the signal is for unlocking, the unlocking unit **46** unlocks the handlebar lock module **42**.

(41) The handlebar lock module **42** is thereby unlocked, and now the handlebar can be turned.

(42) In the present embodiment, the control device **40** is disposed in the space demarcated by the seat rail **30** and the rear frame **20** to be on the outer side of the seat rail **30** and on the inner side of the side cover **32**, so that communication by the communication unit **44** of the control device **40** is not hindered by a metal part, and stability of communication can be improved.

(43) Moreover, providing the control device **40** in the space covered by and in the inner side of the side cover **32** that protects vehicle body parts makes it difficult to remove the control device **40** from the vehicle, and thus secures theft prevention. Furthermore, with the control device **40** provided on the side where the exhaust device **25** is installed, in case of accidental falling, the exhaust device **25** first hits to mitigate an impact on the control device **40**. This improves impact resistance of the control device **40** against accidental falling without improving durability of the control device **40**.

(44) In addition, fixing the control device **40** to the vehicle body frame **11** firmly fixes the control device **40** to improve impact resistance.

(45) As described above, the present embodiment includes the communication unit **44** for communicating with the smart key **43** (device external to the vehicle), and the communication unit **44** is provided in the space below and on the outer side in the vehicle width direction of the seat rail **30**, and is covered by and on the inner side of the side cover **32** (vehicle body cover) that protects the vehicle body parts.

(46) Since the communication unit **44** is provided on a side, in the vehicle width direction, where the exhaust device **25** is provided, below and on the outer side in the vehicle width direction of the seat rail **30**, and in a space covered by and on the inner side of the side cover **32** that protects the vehicle body parts, communication made by the communication unit **44** is not hindered by a metal part, and stable communication can be improved. Moreover, providing the communication unit **44** in the space covered by and in the inner side of the side cover **32** that protects the vehicle body parts makes it difficult to remove the communication unit **44** from the vehicle, and this secures

theft prevention.

(47) In addition, the present embodiment includes the ECU **41** (control unit) that controls a device mounted on the vehicle, and the control device **40** including the communication unit **44** and the ECU **41**.

(48) The control device **40** including the communication unit **44** is provided below and on the outer side in the vehicle width direction of the seat rail and in the space covered by and on the inner side of the side cover **32** that protects the vehicle body parts, so that stability of communication by the communication unit **44** can be improved. Moreover, providing the control device **40** in the space covered by and in the inner side of the side cover **32** that protects vehicle body parts makes it difficult to remove the control device **40** from the vehicle, and thus secures theft prevention.

(49) Furthermore, in the present embodiment, the vehicle includes the handlebar lock module **42** that locks turning of the steering shaft, and the ECU **41** (control unit) determines whether to unlock the handlebar lock module.

(50) According to this configuration, the control device **40** can unlock the handlebar lock module by the ECU **41** making determination to unlock the handlebar lock module.

(51) The present embodiment includes the exhaust device **25** (exhaust mechanism) for exhausting substances produced in a process of operating the power unit **12** (drive device) to the outside of the vehicle, and the communication unit **44** is provided on a side, in the vehicle width direction, where the exhaust device **25** is provided

(52) With the communication unit **44** provided on the side where the exhaust device **25** is installed, in case of accidental falling, the exhaust device **25** first hits to mitigate an impact on the communication unit **44**. This improves impact resistance of the communication unit **44** against falling without improving durability of the control device **40**.

(53) Furthermore, in the present embodiment, the communication unit **44** is attached to the vehicle body frame **11**.

(54) Fixing the communication unit **44** to the vehicle body frame **11** firmly fixes the communication unit **44** to improve impact resistance.

(55) Furthermore, in the present embodiment, the vehicle includes the battery **33** (electric storage device) for operating at least one electrical component provided in the vehicle, and the battery **33** is provided on a side opposite to, in the vehicle width direction with respect to the vehicle center as an axis, the side where the communication unit **44** is provided.

(56) Providing the communication unit **44** on the side opposite to the side where there are many metal parts and where the battery **33** is provided in an easily detachable manner can improve stability of communication and theft prevention without using additional parts.

Modification

(57) Next, an exemplary modification of the present invention will be described.

(58) FIG. **6** is a cross-sectional view illustrating an exemplary modification of the present invention.

(59) As illustrated in FIG. **6**, in the present exemplary modification, a control device **40** including a communication unit **44** is attached to an inner surface of a side cover **32**.

(60) With the control device **40** including the communication unit **44** attached to the inner surface of the side cover **32**, vibration of the vehicle body can be absorbed by the side cover **32**. Thus, the control device **40** including the communication unit **44** can be installed in the vehicle without improving vibration resistance of the control device **40** including the communication unit **44**.

(61) FIG. **7** is a cross-sectional view illustrating another exemplary modification of the present invention.

(62) As illustrated in FIG. **7**, an air cleaner **31** for removing impurities from air taken from outside the vehicle into the vehicle is provided below a seat rail **30**.

(63) In the present exemplary modification, a control device **40** including a communication unit **44** is attached to an outer side in the vehicle width direction of the air cleaner **31**. The control device

40 including the communication unit **44** is attached to be disposed in the outer side in the vehicle width direction of the seat rail **30**.

(64) Since the air cleaner **31** is installed at a place where vibration of the vehicle body is small and is made of a non-metallic material such as resin, stable communication can be maintained without improving vibration resistance of the control device **40** including the communication unit **44** by attaching the control device **40** including the communication unit **44** to the air cleaner **31**.

(65) The present invention is not limited to the configuration of the above embodiment, and can be put into practice in various embodiments without departing from the gist of the invention.

(66) For example, the vehicle according to the present invention is not limited to that uses an engine as the power unit **12**, and can also be applied to a vehicle using an electric motor.

(67) [Configurations Supported by Above Embodiment]

(68) The above embodiment supports the following configurations.

(69) (First configuration) A vehicle including a steering shaft, a vehicle body frame including a head pipe that turnably supports the steering shaft, a main frame that extends from the head pipe to a rear side of the vehicle, and a seat rail that is continuous with the main frame and disposed below a seat on which a rider of the vehicle is seated to fix the seat, and a drive device that generates power for the vehicle, the vehicle including a communication unit for communicating with a device external to the vehicle, where the communication unit is provided below and on an outer side in a vehicle width direction of the seat rail and in a space covered by and on an inner side of a vehicle body cover that protects a vehicle body part.

(70) According to this configuration, since the communication unit is provided below and on the outer side in the vehicle width direction of the seat rail and in the space covered by and on the inner side of the vehicle body cover that protects the vehicle body parts, communication made by the communication unit is not hindered by the metal parts, so that stability of communication can be improved. Moreover, providing the communication unit in the space covered by and in the inner side of the vehicle body cover that protects the vehicle body parts makes it difficult to remove the communication unit from the vehicle, which secures theft prevention.

(71) (Second configuration) The vehicle according to configuration **1**, including a control unit that controls at least a device mounted on the vehicle, and a control device including the communication unit and the control unit.

(72) According to this configuration, the control device including the communication unit is provided below and on the outer side in the vehicle width direction of the seat rail and in the space covered by and on the inner side of the vehicle body cover that protects the vehicle body parts, so that stability of communication by the communication unit can be improved. Moreover, providing the control device in the space covered by and in the inner side of the vehicle body cover that protects the vehicle body parts makes it difficult to remove the control device from the vehicle, which secures theft prevention.

(73) (Third configuration) The vehicle according to the second configuration, including a handlebar lock module that locks turning of the steering shaft, where the control device makes determination to unlock the handlebar lock module.

(74) According to this configuration, the control unit makes determination to unlock the handlebar lock module and the control device can unlock the handlebar lock module.

(75) (Fourth configuration) The vehicle according to any one of the first to third configurations, including an exhaust mechanism for exhausting a substance produced in a process of operating the drive device to an outside of the vehicle, where the communication unit is provided on a side, in the vehicle width direction, where the exhaust mechanism is provided.

(76) According to this configuration, the control device is provided on the side where the exhaust mechanism is installed, and in case of accidental falling, the exhaust mechanism first hits to mitigate an impact on the control device. This improves impact resistance of the control device against accidental falling without improving durability of the control device.

(77) (Fifth configuration) The vehicle according to any one of the first to fourth configurations, where the communication unit is attached to the vehicle body frame.

(78) According to this configuration, fixing the control device to the vehicle body frame firmly fixes the control device to improve impact resistance.

(79) (Sixth configuration) The vehicle according to any one of the first to fourth configurations, where the communication unit is attached to the vehicle body cover.

(80) According to this configuration, the control device including the communication unit is attached to the inner surface of the vehicle body cover, and vibration of the vehicle body can be absorbed by the vehicle body cover. Thus, the control device can be installed in the vehicle without improving vibration resistance of the control device.

(81) (Seventh configuration) The vehicle according to any one of the first to fourth configurations, including an air cleaner that is provided below the seat and removes impurities from air taken from outside the vehicle into an inside of the vehicle, where the communication unit is attached to the air cleaner.

(82) According to this configuration, since the control device is attached to the air cleaner installed at a place where vibration of the vehicle body is small and is made of a non-metallic material such as resin, stable communication can be maintained without improving vibration resistance of the control device itself.

(83) (Eighth configuration) The vehicle according to any one of the first to seventh configurations, including an electric storage device for operating at least one electrical component provided in the vehicle, where the electric storage device is provided on a side opposite to, in the vehicle width direction with respect to the vehicle center as an axis, a side where the communication unit is provided.

(84) According to this configuration, providing the control device on the side opposite to the side where there are many metal parts and where the electric storage device is provided in an easily detachable manner can improve stability of communication, and improve theft prevention without using additional parts.

REFERENCE SIGNS LIST

(85) **10** saddle-ride vehicle **11** vehicle body frame **12** power unit **17** seat **19** front frame **20** rear frame **21** handlebar **23** crankcase **24** cylinder portion **25** exhaust device **26** front fender **27** rear fender **28** step **29** fuel tank **30** seat rail **31** air cleaner **32** side cover **33** battery **40** control device **42** handlebar lock module **43** smart key **44** communication unit **45** determination unit **46** unlocking unit

Claims

1. A vehicle, comprising: a steering, shaft; a vehicle body frame including a head pipe that turnably supports the steering shaft, a main frame that extends from the head pipe to a rear side of the vehicle, and a seat rail that is continuous with the main frame and disposed below a seat on which a rider of the vehicle is seated to fix the seat; a drive device that is configured with an engine or an electric motor generating power for the vehicle; a communicator that receives a signal from a device external to the vehicle and that transmits the signal to a processor controlling another device mounted on the vehicle; and an air cleaner that is for removing impurities from air taken from outside the vehicle into the vehicle that is provided below the seat rail, the air cleaner being made of a non-metallic material, wherein the communicator is provided below and on an outer side in a vehicle width direction of the seat rail and in a space covered by and on an inner side of a vehicle body cover that protects a vehicle body part, and the communicator is attached to the air cleaner and disposed at the outer side of the seat rail in the vehicle width direction.

2. The vehicle according to claim 1, further comprising: a control device including the communicator and the processor.

3. The vehicle according to claim 2, further comprising a handlebar lock module that locks turning of the steering shaft, wherein the processor makes a determination to unlock the handlebar lock module.
 4. The vehicle according to claim 1, wherein the drive device is the engine, and the vehicle further comprises an exhaust mechanism for exhausting a substance produced in a process of operating the engine to an outside of the vehicle, wherein the communicator is provided on a side, in the vehicle width direction, where the exhaust mechanism is provided.
 5. The vehicle according to claim 1, further comprising a battery for operating at least one electrical component provided in the vehicle, wherein the battery is provided on a side opposite to, in the vehicle width direction with respect to a vehicle center as an axis, a side where the communicator is provided.
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